Learning Objective 3.8: Predict and quantify the three ways a real steered array departs from the ideal pattern: grating lobes from element spacing, beam squint from operating away from the phase-set frequency, and the pointing and sidelobe limits of a B-bit phase shifter.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Where the array factor repeats.** The ADALM-PHASER carries 8 patch elements on a 14 mm pitch. In a thinning demonstration only every third element is fed, so the fed elements sit 42 mm apart. Work at the array design centre, 10.3 GHz, where $\lambda = 29.1$ mm.

   (a) In one or two sentences, explain why a grating lobe reaches the same height as the main lobe, while a sidelobe does not.

   (b) With the beam at broadside, find every grating-lobe angle in visible space.

   $\theta_g =$ ☐

   (c) Now steer the beam to $\theta_0 = +20°$. Find every grating lobe that remains in visible space.

   $\theta_g =$ ☐

   (d) A classmate proposes applying a Hann taper to remove the extra beams. Explain in one or two sentences why that does not work.

2. **Choosing the element spacing.** An X-band linear array is being laid out. Use $c = 3 \times 10^8$ m/s throughout.

   (a) State the criterion that keeps grating lobes out of visible space over a scan range $\pm\theta_0$, and explain why half-wavelength spacing is the usual default.

   (b) The array must scan to $\pm 60^\circ$ and operate up to 10.5 GHz. Find the largest usable element spacing in millimetres.

   $d_{\text{max}} = \boxed{}$

   (c) A competing layout uses $d = 20$ mm. At 10.5 GHz, how far off broadside can that array scan before a grating lobe enters visible space?

   $\theta_{0,\text{max}} = \boxed{}$

   (d) The PHASER uses $d = 14$ mm. State its scan limit from the same criterion, and give one sentence on what the design gives up by choosing a spacing this tight.

3. **Beam squint across a band.** An 8-element array with $d = 14$ mm operates from 10.0 to 10.5 GHz. The beam-steering computer sets the element phases for $\theta_0 = 45^\circ$ at the band centre, $f_0 = 10.25$ GHz. Use $c = 3 \times 10^8$ m/s.

   (a) Explain in one or two sentences why a broadside beam does not squint at all.

   (b) Find the beam angle at the low band edge, 10.0 GHz.

   $\theta =$

   (c) Find the beam angle at the high band edge, 10.5 GHz, and state the total angular walk of the beam across the band.

   walk =

   (d) The half-power beamwidth of this array at 45° is $\theta_{\text{HP}} \approx 0.886\lambda/(Nd\cos(\theta_0))$. Express the walk from part (c) as a fraction of a beamwidth, and recommend a fix if the requirement were to hold the walk under 2% of a beamwidth.

   walk / θ_{HP} =

4. **What the phase shifter can actually do.** The ADAR1000 beamformer on the PHASER uses a B-bit phase shifter. Work at the HB100 source frequency, 10.525 GHz, where $\lambda = 28.5$ mm and $d/\lambda = 0.491$.

   (a) Find the phase step (LSB) for $B = 6$ and for the ADAR1000's $B = 7$.

   $B = 6$:

   $B = 7$:

   (b) With $B = 7$, find the pointing step near broadside and at $\theta_0 = 60°$.

   broadside:

   at 60°:

   (c) Use the rule of thumb QSLL $\approx -6B$ dB for $B = 3$ and $B = 7$, and state which sidelobe mechanism dominates in each case for a uniform 8-element array.

   $B = 3$:

   $B = 7$:

   (d) A classmate argues that because the phase shifter resolves only 2.8125°, a pattern null on the PHASER can never be driven deeper than the 21 dB a sweep shows. Explain in two or three sentences why that reasoning is wrong, and state what does set the measured depth.

5. **Reading a sweep and sizing a design.** Each part below reports something a student measured on the PHASER, or asks for a design decision. The array is 8 elements on a 14 mm pitch unless stated otherwise.

   (a) A broadside sweep against the HB100 at 10.525 GHz (λ = 28.5 mm) shows three peaks of equal height, at 0° and ±42°. What effective element spacing does that indicate, and how were the elements fed?

   $d_{\mathrm{eff}} =$ $\boxed{}$

   (b) The array is commanded to +45° with phases set at 10.525 GHz, but the sweep peaks at +47.9° with the source running at 10.025 GHz. Name the effect and confirm the measurement.

   $\theta =$ $\boxed{}$

   (c) At $B = 2$ the sweep shows a lobe near −55° about 7 dB below the peak, while the rule of thumb predicts −12 dB. Explain the discrepancy in one or two sentences.

   (d) A new array must scan to ±50° over 10.0 to 10.5 GHz and hold every sidelobe below −25 dB. State the maximum element spacing and the minimum number of phase bits, and say which of the three defects the bandwidth requirement drives.

   $d_{\max} =$ $\boxed{}$

   $B_{\min} =$ $\boxed{}$

Documentation: ____